

Generation Interconnection System Impact Study Report

For Queue Project Q13-0269

Nashua 138 kV
December 01, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0269. The Interconnection Customer proposes a 820.4 MW natural gas combined cycle facility interconnecting at the Nashua 138 kV substation in the Virginia area. The request was received on September 09, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 137 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$142,321,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Nashua – Dickens 138 kV, the Blue – Black Rock 138 kV loads to 412.3% of its 192 MVA rating. Project DFAX: 100.0%.

For the loss of Dickens – Nashua 138 kV, the Black Rock – Nashua 138 kV loads to 395.5% of its 204 MVA rating. Project DFAX: 100.0%.

For the loss of Nashua – Black Rock Substation 138 kV, the Nashua – Dickens 138 kV loads to 373.6% of its 216 MVA rating. Project DFAX: 100.0%.

Vail – Dickens 138 kV: 365.8% loading (219 MVA) following the outage of the Black Rock – Nashua 138 kV; DFAX = 100.0%.

Vail – N Bristol 138 kV: 182.1% loading (311 MVA) following the outage of the Black Rock – Nashua Substation 138 kV; DFAX = 70.0%.

The Blue – Stockton-Sheldon 138 kV is overloaded to 169.1% of its 230 MVA emergency rating for the loss of the Nashua – Dickens 138 kV. The project contributes 52.2% DFAX (428.0 MW) to this overload.

Sour Springs Switch – Blue 138 kV: 169.0% loading (234 MVA) following the outage of the Dickens Substation – Nahua 138 kV; DFAX = 47.8%.

For the loss of Nashua Substation – Dickens 138 kV, the Sour Springs Switch – Hobart 138 kV loads to 167.0% of its 228 MVA rating. Project DFAX: 47.8%.

Atoka – Stockton-Sheldon 138 kV Line: 161.1% loading (234 MVA) following the outage of the Dickens Substation – Nashua 138 kV; DFAX = 52.2%.

Holtville – Vail 138 kV: 128.0% loading (228 MVA) following the outage of the N Bristol – Vail 138 kV; DFAX = 32.1%.

The Shannon Hil – Holtville 138 kV is overloaded to 118.9% of its 234 MVA emergency rating for the loss of the Vail – N Bristol 138 kV line. The project contributes 32.1% DFAX (263.7 MW) to this overload.

The Biglow Canyon – Shannon Hill 138 kV is overloaded to 115.1% of its 237 MVA emergency rating for the loss of the Vail – North Bristol 138 kV. The project contributes 32.1% DFAX (263.7 MW) to this overload.

Affinity – Batesville 138 kV: 113.4% loading (150 MVA) following the outage of the Kean Mtn And Grassy Crk – Barr Lake 345 kV line; DFAX = 7.3%.

The Michigamme Falls Hy – N Bristol 138 kV is overloaded to 110.0% of its 238 MVA emergency rating for the loss of the Forney Sw – N Bristol 138 kV. The project contributes 19.6% DFAX (160.5 MW) to this overload.

43Rd & S. Land – Cement Plant 138 kV: 109.1% loading (289 MVA) following the outage of the Atoka – John Sevier 345 kV; DFAX = 19.9%.

For the loss of Vail – N Bristol 138 kV, the Biglow Canyon – Cumberland 138 kV loads to 107.5% of its 242 MVA rating. Project DFAX: 32.1%.

For the loss of Barr Lake – New Hope Ap 345 kV, the Parke – Taloga 138 kV Line loads to 106.7% of its 270 MVA rating. Project DFAX: 10.8%.

Sandy Ridge – Coal Creek 138 kV: 105.3% loading (236 MVA) following the outage of the Basilone – Faraday 138 kV; DFAX = 10.3%.

For the loss of Dickens – Nashua 138 kV line, the Coyote – Permac 138 kV loads to 103.2% of its 159 MVA rating. Project DFAX: 14.8%.

For the loss of Basilone – Faraday 138 kV line, the Claypool Hill – Coal Crk 138 kV loads to 103.0% of its 328 MVA rating. Project DFAX: 7.3%.

For the loss of Faraday – Basilone 138 kV line, the Claypool Hill – King 138 kV loads to 102.4% of its 341 MVA rating. Project DFAX: 7.3%.

The Michigamme Falls Hy – Roger 138 kV Line is overloaded to 101.7% of its 184 MVA emergency rating for the loss of the Forney Sw Substation – North Bristol 138 kV. The project contributes 9.5% DFAX (77.8 MW) to this overload.

Kean Mtn And Grassy Creek 345/1kV XFMR: 101.3% loading (471 MVA) following the outage of the Clinch River – Carlton Hills 345 kV; DFAX = 19.1%.

The Kean Mtn And Grassy Creek 138/1 kV TX is overloaded to 101.3% of its 471 MVA emergency rating for the loss of the Clinch Riv – Carlton Hills 345 kV. The project contributes 19.1% DFAX (157.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Blue – Black Rock 138 kV	Rebuild line with higher-capacity conductor	\$5,133,000	100% (\$5,133,000)
Black Rock – Nashua 138 kV	Rebuild line with higher-capacity conductor	\$19,818,000	100% (\$19,818,000)
Nashua – Dickens 138 kV	Rebuild line with higher-capacity conductor	\$4,971,000	100% (\$4,971,000)
Vail – Dickens 138 kV	Rebuild line with higher-capacity conductor	\$8,952,000	100% (\$8,952,000)
Vail – N Bristol 138 kV	Rebuild line with higher-capacity conductor	\$9,131,000	100% (\$9,131,000)
Blue – Stockton-Sheldon 138 kV	Rebuild line with higher-capacity conductor	\$4,078,000	91% (\$3,702,000)
Sour Springs Switch – Blue 138 kV	Rebuild line with higher-capacity conductor	\$4,203,000	90% (\$3,783,000)
Hobart – Sour Springs Switch 138 kV	Rebuild line with higher-capacity conductor	\$15,479,000	90% (\$13,934,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Atoka – Stockton-Sheldon 138 kV	Rebuild line with higher-capacity conductor	\$8,160,000	91% (\$7,407,000)
Holtville – Vail Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$2,966,000	100% (\$2,966,000)
Holtville – Shannon Hill 138 kV line	Rebuild line with higher-capacity conductor	\$13,510,000	100% (\$13,510,000)
Biglow Canyon – Shannon Hill 138 kV	Rebuild line with higher-capacity conductor	\$12,608,000	100% (\$12,608,000)
Affinity – Batesville 138 kV	Rebuild line with higher-capacity conductor	\$40,845,000	33% (\$13,308,000)
Michigamme Falls Hy – N Bristol 138 kV	Reconductor line	\$755,000	81% (\$613,000)
43Rd & S. Land – Cement Plant 138 kV	Reconductor line	\$5,473,000	70% (\$3,829,000)
Biglow Canyon – Cumberland 138 kV	Reconductor line	\$2,038,000	100% (\$2,038,000)
Parke – Taloga 138 kV	Reconductor line	\$3,364,000	44% (\$1,473,000)
Sandy Ridge – Coal Crk 138 kV Line	Reconductor line	\$1,006,000	31% (\$308,000)
Coyote – Permac 138 kV	Reconductor line	\$5,963,000	74% (\$4,409,000)
Claypool Hill – Coal Crk 138 kV	Reconductor line	\$8,595,000	14% (\$1,235,000)
Claypool Hill – King 138 kV Line	Reconductor line	\$1,041,000	14% (\$150,000)
Michigamme Falls Hy – Roger 138 kV Line	Reconductor line	\$1,916,000	100% (\$1,916,000)
Kean Mtn AndGrassy Crk Substation 345/1kV Transformer Bank	Replace transformer with higher-rated unit	\$13,291,000	37% (\$4,917,000)
Kean Mtn And Grassy Crk 138/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,973,000	37% (\$2,210,000)
	Total allocated to project		\$142,321,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.